Four Eggs, N Floors: Is 1–3 Better Than 2–2?

Abstract

With four eggs and N floors, the worst-case number of drops of any reasonable strategy is of order $N^{1/4}$, so the question "1–3 or 2–2?" is really a question about constants. We show that the best 2–2 strategy needs $2\sqrt{2}\,N^{1/4} + O(1) \approx 2.828\,N^{1/4}$ drops, while the best 1–3 strategy needs $4\left(\frac{2}{9}\right)^{1/4} N^{1/4} + O(1) \approx 2.746\,N^{1/4}$ drops. Hence 1–3 **is better** for large N, by a factor of $(9/8)^{1/4} \approx 1.030$ (about 3%). We also check the answer by an independent counting argument and by exact computer calculations. For context, the best possible (unrestricted) four-egg strategy needs about $24^{1/4}N^{1/4} \approx 2.213\,N^{1/4}$ drops.

1 The problem and the ground rules

A building has N floors. We have k identical eggs. There is a number n (the *critical floor*) such that an egg dropped from any floor below n survives undamaged (and can be reused), while an egg dropped from floor n or higher breaks. If the egg survives even from floor N, we say $n = N + 1$. Our job is to determine n, and we measure a strategy by its *worst-case* number of drops over all possible values of n (this is the interpretation "minimize the worst case" from the talk [1]).

Since $n \in \{1, 2, \ldots, N+1\}$, there are $N+1$ possibilities. A drop from floor x splits the possibilities in two: if the egg breaks then $n \le x$, and if it survives then $n > x$.

Notation. Let $D_k(m)$ be the smallest number of drops that always suffices to find n when we have k eggs and m floors we are allowed to test (so there are $m+1$ possibilities for n). Thus $D_k(N)$ is the answer to the original problem. We set $D_k(0) = 0$.

The two strategies. Fix a block size B (we will call it G for the 1–3 strategy and F for the 2–2 strategy). Group the possibilities $n = 1, \ldots, N+1$ into $M = \lceil (N+1)/B \rceil$ consecutive blocks $\{1, \ldots, B\}$, $\{B+1, \ldots, 2B\}$, …; each block has at most B possibilities. These are the "superfloors" of the problem statement.

- **1–3:** Drop one egg from floors $B, 2B, 3B, \ldots$ in turn (one drop per block), until it breaks. This identifies the block containing n using at most $M - 1$ drops. We now have 3 eggs, and the block has at most B possibilities, i.e. $B - 1$ testable floors; solve this with the best 3-egg strategy.
- **2–2:** Use the best 2-egg strategy on the M blocks, i.e. on $M - 1$ testable "superfloors", to find the block containing n. At most two eggs are lost, so we still have 2 eggs, and we solve the block of $B - 1$ testable floors with the best 2-egg strategy.

Therefore the worst-case costs are (at most)

$$T_{13}(N,G) = (M-1) + D_3(G-1), \qquad M = \lceil (N+1)/G \rceil, \tag{1}$$
$$T_{22}(N,F) = D_2(M-1) + D_2(F-1), \qquad M = \lceil (N+1)/F \rceil. \tag{2}$$

Our task: find $\min_G T_{13}(N,G)$ and $\min_F T_{22}(N,F)$, and compare them. Note that in both strategies the "sub-problems" are solved *optimally*; in particular the 2-egg searches use the decreasing steps $x, x-1, x-2, \ldots$ (triangular numbers) discussed in the talk, not constant steps. So we first need to know D_2 and D_3 precisely.

2 How many floors can d drops handle with k eggs?

Rather than computing $D_k(m)$ directly, it is easier to flip the question around: given d drops and k eggs, what is the *largest* number $g(d,k)$ of floors that can be handled?

Lemma 1 (Capacity)**.** *For $d \geq 0$ and $k \geq 0$,*

$$g(d,k) = f(d,k) := \sum_{i=1}^{k} \binom{d}{i}.$$

That is, $D_k(m) \leq d$ if and only if $m \leq f(d,k)$.

Proof. Clearly $g(0,k) = 0$ (with no drops we can only handle $m = 0$ testable floors) and $g(d,0) = 0$ (with no eggs we cannot test anything). Now let $d, k \geq 1$ and suppose a strategy resolves m floors using at most d drops. Its first drop is from some floor $x \in \{1, \ldots, m\}$.

- If the egg breaks, $n \in \{1, \ldots, x\}$: we have $k-1$ eggs and $d-1$ drops left and $x-1$ testable floors, so $x - 1 \leq g(d-1, k-1)$.
- If it survives, $n \in \{x+1, \ldots, m+1\}$: we still have k eggs, $d-1$ drops, and $m-x$ testable floors, so $m - x \leq g(d-1,k)$.

Adding, $m \leq 1 + g(d-1,k-1) + g(d-1,k)$. Conversely, if m equals this bound, drop first from floor $x = 1 + g(d-1,k-1)$; both outcomes are then exactly manageable. Hence

$$g(d,k) = 1 + g(d-1,k-1) + g(d-1,k).$$

Now check that f satisfies the same recursion and boundary values. Clearly $f(0,k) = 0 = f(d,0)$. By Pascal's rule $\binom{d}{i} = \binom{d-1}{i-1} + \binom{d-1}{i}$, so

$$\begin{aligned} f(d,k) &= \sum_{i=1}^{k} \binom{d-1}{i-1} + \sum_{i=1}^{k} \binom{d-1}{i} \\ &= \Big[1 + \sum_{j=1}^{k-1} \binom{d-1}{j}\Big] + f(d-1,k) = 1 + f(d-1,k-1) + f(d-1,k). \end{aligned}$$

By induction on d, $g = f$. (Resolving fewer than $f(d,k)$ floors is of course no harder.) □

Examples.

- $k = 1$: $f(d, 1) = d$. (One egg: go up one floor at a time.)
- $k = 2$: $f(d, 2) = d + \binom{d}{2} = \dfrac{d(d+1)}{2}$, the triangular numbers. For example $N = 105 = \frac{14 \cdot 15}{2}$ needs exactly 14 drops, matching the example in the talk.
- $k = 3$: $f(d, 3) = d + \binom{d}{2} + \binom{d}{3} = \dfrac{d^3 + 5d}{6}$.
- $k = 4$: $f(d, 4) = \dfrac{d^4 - 2d^3 + 11d^2 + 14d}{24}$.

In each case the leading term is $d^k/k!$. The next lemma turns this into an explicit formula for the number of drops.

Lemma 2 (Size of $D_k(m)$)**.** *For all $k \geq 1$ and $m \geq 1$,*

$$\left| D_k(m) - (k!\, m)^{1/k} \right| \leq k.$$

In particular $D_2(m) = \sqrt{2m} + O(1)$, $D_3(m) = (6m)^{1/3} + O(1)$, $D_4(m) = (24m)^{1/4} + O(1)$.

Proof. Write $x = (k!\, m)^{1/k}$ and $D = D_k(m)$; by Lemma 1, D is the least integer d with $f(d, k) \geq m$.

Lower bound on D. Since $\binom{d}{i} \leq d^i/i!$,

$$f(d, k) \leq \sum_{i=1}^{k} \frac{d^i}{i!} \leq \frac{(d+k)^k}{k!}.$$

(Indeed, by the binomial theorem $\frac{(d+k)^k}{k!} = \sum_{i=0}^{k} \frac{d^i}{i!} \cdot \frac{k^{k-i}}{(k-i)!}$, and $\frac{k^j}{j!} \geq 1$ for $0 \leq j \leq k$, so each term with $1 \leq i \leq k$ is at least $d^i/i!$.) Then $m \leq f(D, k) \leq (D+k)^k/k!$ gives $D \geq x - k$.

Upper bound on D. By minimality $f(D-1, k) < m$. If $D - 1 \geq k - 1$, then

$$f(D-1, k) \geq \binom{D-1}{k} = \frac{(D-1)(D-2)\cdots(D-k)}{k!} \geq \frac{(D-k)^k}{k!},$$

so $(D-k)^k < k!\, m$, i.e. $D < x + k$. If $D - 1 < k - 1$ then $D < k \leq x + k$ trivially. □

3 The 2–2 strategy

By (2) and Lemma 2,

$$T_{22}(N, F) = \sqrt{2(M-1)} + \sqrt{2(F-1)} + O(1).$$

Since $|M - 1 - N/F| \leq 1$ and $|\sqrt{a} - \sqrt{b}| \leq \sqrt{|a-b|}$, replacing $M - 1$ by N/F and $F - 1$ by F changes the total by at most an absolute constant. So, uniformly in N and F,

$$T_{22}(N, F) = \sqrt{\frac{2N}{F}} + \sqrt{2F} + O(1). \tag{3}$$

The two terms compete: a large F makes the first stage cheap and the second stage expensive, and vice versa. Let $h(F) = \sqrt{2}\,(N^{1/2}F^{-1/2} + F^{1/2})$. Then

$$h'(F) = \sqrt{2}\Big(-\tfrac{1}{2}N^{1/2}F^{-3/2} + \tfrac{1}{2}F^{-1/2}\Big) = \frac{\sqrt{2}}{2}F^{-3/2}\big(F - N^{1/2}\big),$$

which is negative for $F < \sqrt{N}$ and positive for $F > \sqrt{N}$. So the unique minimum is at $F = \sqrt{N}$, where each term equals $\sqrt{2}\,N^{1/4}$:

$$\min_F T_{22}(N, F) = 2\sqrt{2}\,N^{1/4} + O(1) \approx 2.8284\,N^{1/4}. \tag{4}$$

(There is a quicker way to see the lower bound: by AM–GM, $\sqrt{2N/F}+\sqrt{2F} \geq 2\sqrt{\sqrt{2N/F}\sqrt{2F}} = 2\sqrt{2\sqrt{N}} = 2\sqrt{2}\,N^{1/4}$, with equality iff $F = \sqrt{N}$. Rounding F to an integer costs only $O(1)$.)

4 The 1–3 strategy

By (1), Lemma 2, $M - 1 = N/G + O(1)$, and $|a^{1/3} - b^{1/3}| \leq |a - b|^{1/3}$, uniformly in N and G,

$$T_{13}(N, G) = \frac{N}{G} + (6G)^{1/3} + O(1). \tag{5}$$

Now the superfloor size should be much larger than $\sqrt{N}$: the first egg is only allowed a few drops, but the 3-egg search is very efficient. Following the hint in the problem, write $G = c\,N^{3/4}$ with $c > 0$ to be chosen. Then

$$\frac{N}{G} = \frac{1}{c}\,N^{1/4}, \qquad (6G)^{1/3} = (6c)^{1/3}N^{1/4},$$

so $T_{13} = \phi(c)\,N^{1/4} + O(1)$ with

$$\phi(c) = \frac{1}{c} + 6^{1/3}c^{1/3}.$$

We minimize ϕ over $c > 0$. We have

$$\phi'(c) = -\frac{1}{c^2} + \frac{6^{1/3}}{3}c^{-2/3},$$

and $\phi'(c) = 0$ iff $c^{-4/3} = 6^{1/3}/3$, i.e. $c^{4/3} = 3 \cdot 6^{-1/3}$, i.e.

$$c = \big(3 \cdot 6^{-1/3}\big)^{3/4} = \Big(\frac{27}{6}\Big)^{1/4} = \Big(\frac{9}{2}\Big)^{1/4} \approx 1.4565.$$

This is the only critical point; $\phi(c) \to \infty$ as $c \to 0^+$ (the $1/c$ term) and as $c \to \infty$ (the $c^{1/3}$ term), so it is the global minimum. (Equivalently, $\phi' < 0$ for $c < c_*$ and $\phi' > 0$ for $c > c_*$.)

To evaluate $\phi(c_*)$ note that the critical-point equation $c^{-2} = \frac{6^{1/3}}{3}c^{-2/3}$, multiplied by c, says $\frac{1}{c} = \frac{1}{3}\,6^{1/3}c^{1/3}$: at the optimum the first term of ϕ is exactly one third of the second. So

$$\phi(c_*) = \frac{1}{c_*} + \frac{3}{c_*} = \frac{4}{c_*} = 4\Big(\frac{2}{9}\Big)^{1/4} \approx 2.7464.$$

Thus

$$\min_G T_{13}(N, G) = 4\Big(\frac{2}{9}\Big)^{1/4}N^{1/4} + O(1) \approx 2.7464\,N^{1/4}, \tag{6}$$

attained near $G = (9/2)^{1/4}N^{3/4} \approx 1.4565\,N^{3/4}$. At this choice the first egg uses about $0.687\,N^{1/4}$ drops and the 3-egg stage about $2.060\,N^{1/4}$—a $1:3$ split of the work.

An AM–GM check. We can avoid calculus entirely. Write $\frac{N}{G} + (6G)^{1/3}$ as the sum of four terms: $\frac{N}{G}$ and three copies of $\frac{1}{3}(6G)^{1/3}$. By AM–GM,

$$\frac{N}{G} + 3 \cdot \frac{(6G)^{1/3}}{3} \;\geq\; 4\Big(\frac{N}{G} \cdot \frac{6G}{27}\Big)^{1/4} = 4\Big(\frac{2N}{9}\Big)^{1/4},$$

with equality when $\frac{N}{G} = \frac{1}{3}(6G)^{1/3}$ — the same condition as above. The G's cancel inside the fourth root, which is exactly why the bound is independent of G.

5 The comparison

Theorem 1. *For large N, the best 1–3 strategy beats the best 2–2 strategy:*

$$\min_F T_{22}(N,F) - \min_G T_{13}(N,G) = \Big(2\sqrt{2} - 4\big(\tfrac{2}{9}\big)^{1/4}\Big) N^{1/4} + O(1) \approx 0.082\, N^{1/4},$$

and

$$\frac{\min_F T_{22}}{\min_G T_{13}} \;\longrightarrow\; \Big(\frac{9}{8}\Big)^{1/4} \approx 1.030 \qquad (N \to \infty).$$

Proof. Both minima are given by (4) and (6). It remains to compare the constants $2\sqrt{2}$ and $4(2/9)^{1/4}$. Both are positive, so we may compare their fourth powers:

$$\big(2\sqrt{2}\big)^4 = 64, \qquad \Big(4\big(\tfrac{2}{9}\big)^{1/4}\Big)^4 = 256 \cdot \frac{2}{9} = \frac{512}{9} \approx 56.9.$$

Since $64 > 512/9$, we get $2\sqrt{2} > 4(2/9)^{1/4}$, and the ratio of the fourth powers is $64 \big/ \frac{512}{9} = \frac{9}{8}$. Because the difference of the leading terms grows like a positive constant times $N^{1/4}$ while the error terms are bounded, 1–3 wins for all sufficiently large N. □

So the two "fourth-power constants" are 64 for 2–2 and 512/9 for 1–3; smaller is better.

6 A second derivation: counting floors per drop

Here is a cleaner way to see where 64 and 512/9 come from, and why 1–3 wins. Suppose we are allowed d drops in total. Roughly (ignoring lower-order terms, by Lemma 1) a drops with k eggs can resolve $a^k/k!$ floors, and the two stages of our strategies *multiply*: the number of blocks times the block size is the number of floors.

- **2–2:** spend a drops on the superfloors and $b = d - a$ on the floors inside one superfloor. We resolve about $\frac{a^2}{2} \cdot \frac{b^2}{2} = \frac{a^2 b^2}{4}$ floors. This is maximized at $a = b = d/2$ (by AM–GM or calculus), giving $\frac{d^4}{64}$.

- **1–3:** the first egg resolves a blocks, the 3-egg stage $\frac{b^3}{6}$ floors, so we resolve about $\frac{ab^3}{6}$ floors. With $b = d - a$, $\frac{d}{da}\big[a(d-a)^3\big] = (d-a)^2(d-4a)$, which vanishes at $a = d/4$, $b = 3d/4$, giving $\frac{1}{6} \cdot \frac{d}{4} \cdot \frac{27d^3}{64} = \frac{9d^4}{512}$.

For comparison, the best possible 4-egg strategy resolves $\frac{d^4}{24}$ floors. In summary, d drops can resolve about

$$\frac{d^4}{64}\ (2\text{–}2)\ = 0.0156\,d^4, \qquad \frac{9d^4}{512}\ (1\text{–}3)\ = 0.0176\,d^4, \qquad \frac{d^4}{24}\ (\text{optimal})\ = 0.0417\,d^4.$$

Setting this equal to N and solving for d gives $d \approx (64N)^{1/4} = 2\sqrt{2}\,N^{1/4}$, $d \approx (512N/9)^{1/4} = 4(2/9)^{1/4}N^{1/4}$, and $d \approx (24N)^{1/4}$, in agreement with (4), (6) and Lemma 2. The optimal split $a = d/4$, $b = 3d/4$ is also the $1:3$ work split found in Section 4.

Why does 1–3 win? The count of floors handled is a product of two stage capacities, and 2–2 wastes some of the cleverness available: the best way to use the eggs is one in which each egg's drops are "worth" as much as possible, and an egg is most valuable when there are many eggs behind it. A 1–3 strategy puts a *single* egg in the expensive linear first stage and leaves the remaining three for the powerful cubic stage; 2–2 puts two eggs into a quadratic stage twice. In fact the true optimal 4-egg strategy has exactly the "drop one egg; if it breaks, we have 3 eggs" structure, with first-egg steps that shrink as we go up (like the $x, x-1, x-2, \ldots$ trick for two eggs); 1–3 with constant steps G is a uniform approximation of it. This is an intuition rather than part of the proof—the proof is the comparison of constants above.

7 Numerical check

Using the exact formulas (1) and (2) with $D_k(m)$ computed exactly from Lemma 1 (the smallest d with $\sum_{i=1}^{k}\binom{d}{i} \geq m$), and minimizing over the block size, we get the following (the minimizing G or F is not always unique; one minimizer is shown).

N	best G	1–3 drops	best F	2–2 drops	optimal $D_4(N)$
10^2	26	8	7	8	8
10^3	167	15	22	15	13
10^4	1 112	27	65	28	23
10^6	41 667	86	816	89	71
10^8	1 389 710	274	8 488	283	222
10^{10}	44 873 824	868	96 394	894	701
10^{12}	1 419 036 467	2 746	984 436	2 828	2 214
prediction (drops/$N^{1/4}$)		2.7464		2.8284	2.2134

At $N = 10^{12}$ we have $N^{1/4} = 1000$, so the drop counts are essentially 1000 times the predicted constants. The best G is about $1.42\,N^{3/4}$ (predicted $1.4565\,N^{3/4}$) and the best F is about $\sqrt{N}$, as predicted. For small N the $O(1)$ terms dominate and the two strategies tie. We also computed, for *every* $N \leq 6000$, the best 1–3 and best 2–2 costs: 1–3 was never worse than 2–2, strictly better for 3214 of these values of N, and tied for the other 2786. (The tabulated costs are those of the stated strategies; the last column is the true optimum over all strategies, from Lemma 1.)

8 Conclusion

With 4 eggs and N floors, the best 2–2 approach takes about $2\sqrt{2}\,N^{1/4} \approx 2.83\,N^{1/4}$ drops (superfloors of size $F \approx \sqrt{N}$), while the best 1–3 approach takes about $4(2/9)^{1/4}N^{1/4} \approx$

$2.75\,N^{1/4}$ drops (superfloors of size $G \approx (9/2)^{1/4}N^{3/4}$). The 1–3 approach is better, by a factor $(9/8)^{1/4} \approx 1.03$. Neither matches the unrestricted optimum $24^{1/4}N^{1/4} \approx 2.21\,N^{1/4}$, which needs the non-uniform spacing of the first egg's drops.

References

[1] S. J. Miller, *Egg Drop Mathematics: It IS all its cracked up to be!*, talk slides, Williams College, https://web.williams.edu/Mathematics/sjmiller/public_html/math/talks/eggdropmath_Hampshire28Jul2026.pdf; video: https://youtu.be/98cpdt2EsCM.

[2] *Egg drop problems*, paper in the Fibonacci Quarterly (see the link for authors), https://web.williams.edu/Mathematics/sjmiller/public_html/math/papers/EggDropProblems_FibonacciQuarterly.pdf.